Homework 3

Incentive-Centered Design

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Acknowledgements

1. Collaborators: Swathi Narashiman (I) and Aravind Ramana V.
2. LLMs referred: ChatGPT, Claude, Gemini.

Solutions

1. Myerson's lemma states that if players' utilities are of the form

$$u_i(v_i, \alpha, p_i) = v_i\alpha - p_i$$

where $v_i \in \mathbb{R}_{\geq 0}$ is the private valuation of player i, α is the allocation of player i, and p_i is the payment imposed on player i, then a given allocation rule $\mathbf{x} : \mathbb{R}^n \to \mathbb{R}^n$ is implementable in dominant strategies if and only if $\mathbf{x}$ is monotone. The goal of this problem is to generalize the necessity part of this lemma.

(a) (10 Points) Suppose that the players' utilities are of the form

$$u_i(v_i, \alpha, p_i) = f(v_i, \alpha) - p_i$$

and that f satisfies the following properties:

(i) $f(v_i, \cdot)$ is nondecreasing for every v_i, and
(ii) for every $0 \leq y < z$ and every $0 \leq \alpha < \beta$,

$$f(z, \alpha) - f(y, \alpha) < f(z, \beta) - f(y, \beta).$$

Show that the necessity part of Myerson's lemma continues to hold. I.e., show that if an allocation rule is implementable in dominant strategies, then it is necessarily monotone.

Hint: Follow the same sequence of arguments as in the proof of Myerson's lemma.

(b) (5 Points) Verify that $f(v_i, \alpha) = v_i\alpha$ (which is usually used) satisfies the two aforementioned properties.

(c) (5 Points) Construct a function f that is not of the form $f(v, \alpha) = kv\alpha$, $k \in \mathbb{R}$, but still satisfies the two aforementioned properties.

Remark: By imposing another mild condition on f, I believe the sufficiency part can also be generalized. Might be fun to try out when you're bored.

Solution:

(a) For player i, let's take two values for bids y and z such that $y < z$. Let the bids of other players be denoted by b_{-i}. Let us define the allocations α and β as follows:

$$\alpha = x_i(y, b_{-i}) \quad ; \quad \beta = x_i(z, b_{-i})$$

Truthfullness at y

$$\implies f(y, \alpha) - p_i(y) \geq f(y, \beta) - p_i(z)$$

Truthfullness at z

$$\implies f(z, \beta) - p_i(z) \geq f(z, \alpha) - p_i(y)$$

Adding both the inequalities, we get,

$$f(y,\alpha) + f(z,\beta) \geq f(y,\beta) + f(z,\alpha)$$

$$\implies f(z,\beta) - f(y,\beta) \geq f(z,\alpha) - f(y,\alpha) \tag{1}$$

Let us assume that $\beta < \alpha$, then by the property (ii) in the question we have,

$$f(z,\beta) - f(y,\beta) < f(z,\alpha) - f(y,\alpha)$$

But this contradictis (1) , hence the assumption is wrong. Therefore,

$$\beta > \alpha$$

$$\implies x_i(z, b_{-i}) > x_i(y, b_{-i})$$

Hence it is necessarily monotone.

(b) Consider $f(v_i, \alpha) = v_i\alpha$,

(i) For a fixed v_i, for $\alpha < \beta$

$$f(v_i,\alpha) = v_i\alpha < v_i\beta = f(v_i,\beta)$$

Hence, $f(v_i, .)$ is non-decreasing for every v_i.

(ii) Let $0 \leq y < z$ and $0 \leq \alpha < \beta$, then

$$f(z,\alpha) - f(y,\alpha) = (z-y)\alpha < (z-y)\beta = f(z,\beta) - f(y,\beta)$$

Hence, both the proeperties hold.

(c) Let our constructed function be,

$$f(v_i,\alpha) = v_i\alpha^2$$

check property (i), for $\alpha < \beta$

$$f(v_i,\alpha) = v_i\alpha^2 < v_i\beta^2 = f(v_i,\beta)$$

check property (ii), for every $0 \leq y < z$ and $0 \leq \alpha < \beta$,

$$f(z,\alpha) - f(y,\alpha) = (z-y)\alpha^2 < (z-y)\beta^2 = f(z,\beta) - f(y,\beta)$$

Hence, the constructed function satisfies both the properties.

2. (10 Points) When proving Myerson's lemma in class, we assumed that the allocation function is differentiable. This problem deals with an extension wherein the allocation function is not differentiable.

   Consider a single-item auction wherein the seller wishes to allocate the item to the player who values it the most. Hence, the allocation function is

$$x_i(b_i, \mathbf{b}_{-i}) = \begin{cases} 1 & \text{if} \quad b_i > \max\{\mathbf{b}_{-i}\}, \\ 0 & \text{Otherwise.} \end{cases} \tag{2}$$

   Using Myerson's lemma, derive the payment rule $\mathbf{p}$ that would make $(\mathbf{x}, \mathbf{p})$ DSIC. Assume boundary condition $p_i(b_i = 0, \mathbf{b}_{-i}) = 0$ for all i and all $\mathbf{b}_{-i}$.

   Hint: Start from equation (8) of the proof of Myerson's lemma (Lecture 20, Slide 10).

   Solution:
   Suppose there exists a payment rule p such that (x, p) is a DSIC mechanism. The allocation rule is defined as

$$x_i(b_i, \mathbf{b}_{-i}) = \begin{cases} 1 & \text{if } b_i > \max\{\mathbf{b}_{-i}\}, \\ 0 & \text{otherwise.} \end{cases} \tag{3}$$

   Let $y < \max\{\mathbf{b}_{-i}\} < z$. From Equation (8) in the proof, we have

$$y\big[x_i(z, \mathbf{b}_{-i}) - x_i(y, \mathbf{b}_{-i})\big] \;\le\; p_i(z, \mathbf{b}_{-i}) - p_i(y, \mathbf{b}_{-i}) \;\le\; z\big[x_i(z, \mathbf{b}_{-i}) - x_i(y, \mathbf{b}_{-i})\big]. \tag{4}$$

   We also have from (3),

$$x_i(y) = 0 \qquad ; \quad x_i(z) = 1$$

   Then, (4) simplifies as

$$y \le p_i(z) - p_i(y) \le z$$

   We know:

$$p_i(0, b_{-i}) = 0$$

   Also, for any $y < \max\{\mathbf{b}_{-i}\}$, the player loses, so:

$$p_i(y, b_{-i}) = 0$$

   So the inequality becomes:

$$y \le p_i(z, b_{-i}) \le z \quad \text{for all } y < \max\{\mathbf{b}_{-i}\} < z$$

   This must hold for all $y < \max\{\mathbf{b}_{-i}\}$. Taking the limit $y \to \max\{\mathbf{b}_{-i}\}^-$:

$$p_i(z, b_{-i}) \ge \max\{\mathbf{b}_{-i}\}$$

   So we get:

$$\max\{\mathbf{b}_{-i}\} \le p_i(z, b_{-i}) \le z$$

   By DSIC, payments must be minimal for winning types, hence:

$$p_i(z, b_{-i}) = \max\{\mathbf{b}_{-i}\}$$

$$p_i(b_i, \mathbf{b}_{-i}) = \begin{cases} 0 & \text{if } b_i \le \max\{\mathbf{b}_{-i}\} \\ \max\{\mathbf{b}_{-i}\} & \text{if } b_i > \max\{\mathbf{b}_{-i}\} \end{cases}$$

3. (3+3+4=10 Points) Compute the revenue-maximizing price when there is only one potential buyer, a single item, and the buyer's valuation density f is

(i) $f(v) = \begin{cases} 1 & 0 \le v \le 1, \\ 0 & \text{else.} \end{cases}$

(ii) $f(v) = \begin{cases} 4v & 0 \le v \le \frac{1}{2}, \\ 4 - 4v & \frac{1}{2} \le v \le 1. \end{cases}$

(iii) $f(v) = \begin{cases} \frac{\pi}{2}\sin(\pi v) & 0 \le v \le 1 \\ 0 & \text{else.} \end{cases}$

In each case, before even you begin solving, use qualitative arguments to guess the range in which the optimal price should lie (greater than, less than, or equal to $\frac{1}{2}$). Then solve to find the exact price.

Solution:

Revenue is given by,

$$R(v) = v(1 - F(v))$$

Intuition . If f(v) is high near 1/2, many buyers have moderate values. So setting price slightly below where f(v) is high increases sale probability enough to maximize revenue. If PDF is symmetric, then optimal occurs close to half 1/2.

From Myerson's formula, the revenue maximizing price is attained whenever the virtual valuation function becomes zero, i.e,

$$\varphi(v) = v - \frac{1 - F(v)}{f(v)} = 0 \tag{5}$$

where, f(.) is the PDF and F(.) is the CDF of the buyer's valuation.

(i) $f(v) = \begin{cases} 1 & 0 \le v \le 1, \\ 0 & \text{else.} \end{cases}$; $F(v) = \begin{cases} v & 0 \le v \le 1, \\ 1 & \text{else.} \end{cases}$

Since,uniform intutively optimal should lie exactly at 1/2. From (5),

$$2v - 1 = 0 \implies v = \frac{1}{2}$$

Hence, intuition is right.

(ii) $f(v) = \begin{cases} 4v & 0 \le v \le \frac{1}{2}, \\ 4 - 4v & \frac{1}{2} \le v \le 1. \end{cases}$; $F(v) = \begin{cases} 2v^2 & 0 \le v \le \frac{1}{2}, \\ 1 - 2(1-v)^2 & \frac{1}{2} \le v \le 1, \\ 1 & otherwise \end{cases}$

Since PDF peaks at 1/2, optimal price should be lower than 1/2. From (5), for $0 \le v \le \frac{1}{2}$,

$$v - \frac{1 - 2v^2}{4v} = 0 \implies v = \frac{1}{\sqrt{6}}$$

for $\frac{1}{2} \le v \le 1$.

$$v - \frac{2(1-v)^2}{4 - 4v} = 0 \implies v = \frac{1}{3} or\ 1$$

Since, $\frac{1}{3} < \frac{1}{2}$, and 1 gives a revenue of 0, we reject these solutions, and take $\frac{1}{\sqrt{6}}$ as the revenue maximizing price. Hence, intuition is right.

(iii) $f(v) = \begin{cases} \frac{\pi}{2}\sin(\pi v) & 0 \le v \le 1 \\ 0 & \text{else.} \end{cases}$; $F(v) = \begin{cases} \frac{1-\cos(\pi v)}{2} & 0 \le v \le 1 \\ 1 & \text{else.} \end{cases}$ Here again, since density is peaking at 1/2, optimal price is inuitively lower than 1/2.

From (5),

$$v - \frac{\frac{1+\cos(\pi v)}{2}}{\frac{\pi}{2}\sin(\pi v)} = 0 \implies \cot(\frac{\pi v}{2}) = \pi v$$

Using desmos to solve the above gives,

$$v = 0.41589$$

Hence, intuition is right.

4. (5 Points) Which of the following probability distributions are regular? Justify your answer.

   (i) $Unif([a,b])$.

   (ii) $Exp(\lambda)$.

Solution

(i) Unif$([a,b])$**.**
Let $v \sim \text{Unif}([a,b])$. Then for $v \in [a,b]$,

$$F(v) = \frac{v-a}{b-a}, \qquad f(v) = \frac{1}{b-a}.$$

Substituting into the definition of the virtual valuation function, we obtain

$$\varphi(v) = v - \frac{1-F(v)}{f(v)} = v - \frac{\frac{b-v}{b-a}}{\frac{1}{b-a}} = v - (b-v) = 2v - b.$$

Since φ is differentiable on (a,b) with derivative

$$\frac{d\varphi}{dv} = 2 > 0,$$

it follows that φ is strictly increasing on $[a,b]$. In particular, φ is nondecreasing, and hence the uniform distribution Unif$([a,b])$ is *regular*.

(ii) Exp(λ)**.**
Let $v \sim \text{Exp}(\lambda)$ with parameter $\lambda > 0$. Then for $v \geq 0$,

$$F(v) = 1 - e^{-\lambda v}, \qquad f(v) = \lambda e^{-\lambda v}.$$

Substituting into the definition of the virtual valuation function, we obtain

$$\varphi(v) = v - \frac{1-F(v)}{f(v)} = v - \frac{e^{-\lambda v}}{\lambda e^{-\lambda v}} = v - \frac{1}{\lambda}.$$

Since φ is differentiable on $(0,\infty)$ with derivative

$$\frac{d\varphi}{dv} = 1 > 0,$$

it follows that φ is strictly increasing on $[0,\infty)$. In particular, φ is nondecreasing, and hence the exponential distribution Exp(λ) is *regular*.

5. (10 Points) A seller wishing to maximize revenue has a single indivisible item and two prospective buyers. Buyer 1 has valuation density $Unif([0,10])$ for the item and Buyer 2 has valuation density $Exp(2)$. Compute the auction that the seller should employ. I.e., for each possible bid profile of the players, determine who (if any) wins the item and at what price. Justify your answer using Myerson's lemma.

Solution:
For buyer 1,

$$f(v) = \begin{cases} \frac{1}{10} & 0 \le v \le 10, \\ 0 & otherwise. \end{cases} \quad ; \quad F(v) = \begin{cases} \frac{v}{10} & 0 \le v \le 10, \\ 1 & otherwise \end{cases}$$

Therefore, the virtual valuation is given by,

$$\varphi_1(v) = v - \frac{1 - F(v)}{f(v)} = 2v - 10 \tag{6}$$

For buyer 2,

$$f(v) = 2e^{-2v} \quad ; \quad F(v) = 1 - e^{-2v}$$

Therefore, the virtual valuation is given by,

$$\varphi_2(v) = v - \frac{1 - F(v)}{f(v)} = v - \frac{1}{2} \tag{7}$$

By Myerson's lemma, the revenue-maximizing auction allocates the item to the buyer with the highest non-negative virtual valuation. Payments are determined by the critical value (threshold bid) at which the winner would still win.

We donot make a sale if both the valuations are negative, from (6) and (7)

$$No\ sale\ if\ , b_1 < 5\ and\ b_2 < \frac{1}{2}$$

Now buyer 1 wins if his virtual valuation (positive) is higher,

$$2b_1 - 10 \ge b_2 - \frac{1}{2}\ \ and\ \ b_1 \ge 5 \tag{8}$$

Buyer 2 wins if his virtual valuation (positive) is higher,

$$b_2 - \frac{1}{2} \ge 2b_1 - 10\ \ and\ \ b_2 \ge \frac{1}{2} \tag{9}$$

Solving the critical points of the inequalities (8) and (9) and taking account of the minimum values of bids we get,

$$2p_1 - 10 = b_2 - \frac{1}{2} \implies p_1 = \frac{b_2 + 9.5}{2}\ and\ p_1 \ge 5$$

$$p_1 = \max(5, \frac{b_2 + 9.5}{2})$$

Similarly,

$$p_2 - \frac{1}{2} = 2b_1 - 10 \implies p_2 = 2b_1 - 9.5\ and\ p_2 \ge \frac{1}{2}$$

$$p_2 = \max(\frac{1}{2}, 2b_1 - 9.5)$$

Hence, the auction can be mathematically represented as,

$$(x_1, x_2, p_1, p_2) = \begin{cases} (1, 0,\ \max\left(5,\ \frac{b_2+9.5}{2}\right),\ 0) & \text{if } 2b_1 - 10 \ge b_2 - \frac{1}{2} \text{ and } b_1 \ge 5, \\ (0, 1,\ 0,\ \max\left(\frac{1}{2},\ 2b_1 - 9.5\right)) & \text{if } b_2 - \frac{1}{2} \ge 2b_1 - 10 \text{ and } b_2 \ge \frac{1}{2}, \\ (0, 0,\ 0,\ 0) & \text{if } b_1 < 5 \text{ and } b_2 < \frac{1}{2}. \end{cases}$$

Hence, the derived mechanism is the optimal auction by Myerson's lemma.

6. (5 Points) Consider a single-item auction with n buyers. The buyers' valuations are independent and identically distributed random variables with F being the cumulative distribution function of buyer 1's valuation. Assume that F is regular. Prove that of all DSIC auctions that allocate the item, the Vickrey auction maximizes revenue.

Remark: Recall that we appealed to this result in proving a result about prior-free auctions.

Solution

We need to prove that among all DSIC auctions that **always** allocate the item, the Vickrey (second-price) auction maximizes expected revenue.

Let $v_1, \ldots, v_n \overset{\text{i.i.d.}}{\sim} F$, where F is regular. For any DSIC mechanism (x, p), Myerson's lemma implies that the expected revenue equals the expected virtual surplus:

$$\bar{R} = \mathbb{E}_{\mathbf{v}}\left[\sum_{i=1}^{n} \varphi(v_i)\, x_i(\mathbf{v})\right],$$

where

$$\varphi(v) = v - \frac{1 - F(v)}{f(v)}$$

is the (common) virtual valuation function.

Claim. Among all DSIC mechanisms satisfying

$$\sum_{i=1}^{n} x_i(\mathbf{v}) = 1 \quad \text{for all } \mathbf{v},$$

the Vickrey auction maximizes $\bar{R}$.

Proof. Since F is regular, the virtual valuation function φ is nondecreasing.

Fix a valuation profile $\mathbf{v}$. Consider the problem of maximizing

$$\sum_{i=1}^{n} \varphi(v_i)\, x_i(\mathbf{v})$$

subject to feasibility:

$$x_i(\mathbf{v}) \geq 0, \quad \sum_{i=1}^{n} x_i(\mathbf{v}) = 1.$$

This is a linear function in x, so the maximum is achieved by assigning all weight to a buyer with the highest virtual valuation:

$$x_i^*(\mathbf{v}) = \mathbf{1}\left[i \in \arg\max_j \varphi(v_j)\right].$$

Since φ is nondecreasing, any maximizer of $\varphi(v_j)$ is also a maximizer of v_j. Hence,

$$x_i^*(\mathbf{v}) = 1 \quad \text{iff} \quad v_i \in \arg\max_j v_j,$$

which is exactly the allocation rule of the Vickrey auction (up to tie-breaking).

Since the objective is maximized pointwise for every $\mathbf{v}$, it follows that it is maximized in expectation:

$$\bar{R}(x) = \mathbb{E}\left[\sum_i \varphi(v_i)\, x_i(\mathbf{v})\right] \leq \mathbb{E}\left[\sum_i \varphi(v_i)\, x_i^*(\mathbf{v})\right] = \bar{R}(x^*).$$

Finally, the Vickrey auction is DSIC and implements the allocation rule x^*, hence it achieves this maximum.

□

7. (5 Points) Consider a multi-parameter environment wherein the valuations of all players are nonnegative for all alternatives. In the lecture, we showed that the VCG mechanism is DSIC, IR, and efficient. Prove that the VCG mechanism is weakly budget-balanced.

Solution.

Under the VCG framework, the chosen outcome maximizes the total reported value:

$$\omega^*(\mathbf{b}) = \arg\max_{\omega\in\Omega} \sum_{i=1}^{n} b_i(\omega).$$

The payment charged to player i is given by

$$p_i(\mathbf{b}) = h_i(\mathbf{b}_{-i}) - \sum_{j\neq i} b_j(\omega^*(\mathbf{b}))\,,$$

where h_i depends only on the bids of players other than i. In the Clarke pivot mechanism, this function is chosen as

$$h_i(\mathbf{b}_{-i}) = \max_{\omega\in\Omega} \sum_{j\neq i} b_j(\omega).$$

Claim. The VCG mechanism is weakly budget-balanced, i.e.,

$$\sum_{i=1}^{n} p_i(\mathbf{b}) \geq 0 \quad \text{for all } \mathbf{b}.$$

Proof. Summing the payments across all players:

$$\sum_{i=1}^{n} p_i(\mathbf{b}) = \sum_{i=1}^{n} h_i(\mathbf{b}_{-i}) - \sum_{i=1}^{n}\sum_{j\neq i} b_j(\omega^*(\mathbf{b}))\,.$$

Observe that in the double sum, each term $b_j(\omega^*)$ appears exactly $n-1$ times (once for every $i \neq j$). Hence,

$$\sum_{i=1}^{n} p_i(\mathbf{b}) = \sum_{i=1}^{n} h_i(\mathbf{b}_{-i}) - (n-1)\sum_{j=1}^{n} b_j(\omega^*(\mathbf{b}))\,.$$

Now, using the definition of h_i, we have

$$h_i(\mathbf{b}_{-i}) \geq \sum_{j\neq i} b_j(\omega^*(\mathbf{b})) = \sum_{j=1}^{n} b_j(\omega^*(\mathbf{b})) - b_i(\omega^*(\mathbf{b}))\,.$$

Summing this inequality over all i:

$$\sum_{i=1}^{n} h_i(\mathbf{b}_{-i}) \geq n\sum_{j=1}^{n} b_j(\omega^*) - \sum_{i=1}^{n} b_i(\omega^*) = (n-1)\sum_{j=1}^{n} b_j(\omega^*).$$

Substituting back, we obtain

$$\sum_{i=1}^{n} p_i(\mathbf{b}) \geq (n-1)\sum_{j=1}^{n} b_j(\omega^*) - (n-1)\sum_{j=1}^{n} b_j(\omega^*) = 0.$$

Thus, the total payment collected is nonnegative for all bid profiles, establishing weak budget balance. □

8. (5+10+5+5=25 Points) The VCG mechanism that we saw in class assumed that the valuations of all players are nonnegative for every outcome. The first part of this problem shows that the incentive compatibility of the VCG mechanism continues to hold even when valuations are negative. The later parts of the problem applies the result of the first part to establish a fundamental impossibility result pertaining to bilateral trade, viz., efficient bilateral trade cannot occur without external subsidy — a result that lies at the heart of tensions in international trade.

(i) Let Ω be a finite set of outcomes and denote by $v_i(\omega) \in \mathbb{R}$ the valuation of player i for outcome $\omega \in \Omega$. The valuation can be any real number; not just nonnegative. Denote by $\mathbf{b}_i$ the bid of player i, $i \in \{1, \ldots, n\}$. Show that if the social planner employs the decision rule

$$\omega^*(\mathbf{b}_1, \ldots, \mathbf{b}_n) = \arg\max_{\omega \in \Omega} \sum_{i=1}^{n} b_i(\omega) \tag{10}$$

and payment rule

$$p_i(\mathbf{b}_i, \mathbf{b}_{-i}) = f_i(\mathbf{b}_{-i}) - \sum_{j \neq i} b_j(\omega^*(\mathbf{b}_1, \ldots, \mathbf{b}_n)) \tag{11}$$

for all $i \in \{1, \ldots, n\}$, where f_i is any arbitrary function, then truthful bidding by all players is a dominant strategy. Also show that at equilibrium, the outcome chosen by the social planner maximizes social welfare.

Remark: Conversely, every DSIC mechanism has a payment rule that is of the form (11) for some f_is.

(ii) Two parties are interested in trading an indivisible good. The seller values the good at v_s and the buyer values the good at v_b. The valuations are private to the players. There are two possible outcomes: (i) trade occurs (Y), or (ii) trade does not occur (N). The valuations of the players for all outcomes are as specified in the following table.

	Y	N
Seller	$-v_s$	0
Buyer	v_b	0

A third person, say a market maker, wishes to implement the following outcome function:

- (Efficiency) If $v_b \geq v_s$, outcome Y must be chosen, and if $v_b < v_s$, outcome N must be chosen. I.e., the good must find itself in the hands of the person who values it the most.

Design a DSIC mechanism which implements the aforementioned outcome function. The mechanism must also satisfy the following two properties:

(a) (Individual Rationality) Both parties must obtain a nonnegative utility regardless of what the outcome is,

(b) In the event that no trade occurs, both parties pay 0.

(iii) Does the mechanism designed in Part (ii) satisfy budget balance?

(iv) Let's relax the requirement that both parties pay 0 in the event that no trade occurs. Can we now design a mechanism that satisfies budget balance? Justify.

Solution:

(i) **Truthful bidding is a dominant strategy.**

Fix a player i with true valuation $v_i(\cdot)$, and assume all other players report truthfully. Under the mechanism defined in (5)–(6), the utility of player i when submitting a bid b_i is

$$u_i = v_i(\omega^*) - p_i = v_i(\omega^*) - f_i(\mathbf{b}_{-i}) + \sum_{j \neq i} b_j(\omega^*).$$

Since $f_i(\mathbf{b}_{-i})$ does not depend on b_i, maximizing u_i reduces to maximizing

$$v_i(\omega^*) + \sum_{j \neq i} b_j(\omega^*).$$

The mechanism selects

$$\omega^* = \arg\max_{\omega} \sum_{k} b_k(\omega).$$

If player i reports truthfully, i.e. $b_i = v_i$, then

$$\omega^* = \arg\max_{\omega} \left(v_i(\omega) + \sum_{j \neq i} b_j(\omega) \right),$$

which directly maximizes the expression relevant for player i's utility.
Any deviation $b_i \neq v_i$ may result in a different outcome ω^{**} such that

$$v_i(\omega^{**}) + \sum_{j \neq i} b_j(\omega^{**}) \;\leq\; v_i(\omega^*) + \sum_{j \neq i} b_j(\omega^*).$$

Thus, truthful reporting yields the highest utility regardless of others' bids, establishing that it is a *dominant strategy*.

Efficiency. When all players bid truthfully ($b_i = v_i$ for all i), the mechanism chooses

$$\omega^* = \arg\max_{\omega} \sum_i v_i(\omega),$$

which maximizes total social welfare.

(ii) **Designing a DSIC and individually rational mechanism.**
We consider a bilateral trade setting with two agents: a seller s and a buyer b. The outcome space is $\Omega = \{Y, N\}$, representing trade and no trade.
The valuation structure is:

$$v_s(Y) = -v_s, \quad v_s(N) = 0, \qquad v_b(Y) = v_b, \quad v_b(N) = 0,$$

where v_s and v_b are privately known.
Efficiency requires that trade occurs (Y) if and only if $v_b \geq v_s$.

We adopt the VCG framework:

$$\omega^*(b_s, b_b) = \arg \max_{\omega \in \{Y,N\}} \big(b_s(\omega) + b_b(\omega)\big),$$

with ties resolved in favor of trade (Y).
Payments take the form:

$$p_s = f_s(b_b) - b_b(\omega^*), \qquad p_b = f_b(b_s) - b_s(\omega^*),$$

where f_s, f_b depend only on the other agent's report. By part (i), any such mechanism is DSIC.

We impose two conditions:
(a) **Individual Rationality:** $u_s \geq 0$, $u_b \geq 0$
(b) **No payments under no trade:** if $\omega^* = N$, then $p_s = p_b = 0$
Under $\omega^* = N$, we have $b_s(N) = b_b(N) = 0$, hence:

$$p_s = f_s(b_b), \quad p_b = f_b(b_s).$$

Condition (b) implies:

$$f_s(b_b) = 0, \quad f_b(b_s) = 0,$$

so we must choose $f_s \equiv 0$ and $f_b \equiv 0$.
Thus the payments simplify to:

$$p_s = -\,b_b(\omega^*), \qquad p_b = -\,b_s(\omega^*).$$

Case 1: $v_b \geq v_s$ **(trade occurs).**

$$p_s = -v_b, \qquad p_b = v_s.$$

Utilities:

$$u_s = -v_s - (-v_b) = v_b - v_s \geq 0, \qquad u_b = v_b - v_s \geq 0.$$

Case 2: $v_b < v_s$ **(no trade).**

$$p_s = p_b = 0, \qquad u_s = 0, \quad u_b = 0.$$

Thus, both individual rationality and the zero-payment condition hold.

Each agent reports their valuation. Trade occurs if $b_b \geq b_s$. In that case, the buyer pays b_s, and the seller receives b_b. If $b_b < b_s$, no trade occurs and no payments are made.

(iii) **Budget balance.**

If trade occurs:

$$p_s + p_b = -v_b + v_s = -(v_b - v_s) \leq 0,$$

with strict inequality when $v_b > v_s$.

If no trade occurs, both payments are zero.

Hence, the mechanism never generates a surplus and may require external subsidy. Therefore, it is **not budget-balanced**.

(iv) **Impossibility result.**

Even if we relax the requirement that payments vanish under no trade, it is impossible to simultaneously achieve:

- DSIC,
- ex-post individual rationality,
- efficiency,
- (weak) budget balance.

This follows from the *Myerson–Satterthwaite theorem*.

Intuitively, ensuring truthful reporting requires leaving informational rents to both agents, and these rents exceed the gains from trade. As a result, any mechanism satisfying the above properties must incur a deficit in expectation.